

Payment Ecosystem

Complete Tutorial

How money moves between banks, merchants, and customers.

Written in simple A1-level English. Read this before your CIBPay interview.

This tutorial explains the complete payment ecosystem: who are the players, how money moves from one bank to another, and what role a payment system like CIBPay plays in this process. You will learn about acquiring banks, issuing banks, card networks, settlement, clearing, and much more. Every explanation is in simple English so you can understand and remember easily.

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CHAPTER 1

Key Players in the Payment Ecosystem

When a customer buys something online and pays with a card, many different organizations work together to make this happen. Think of it like a team: each player has a different job, but they all work together to complete the payment. Let us meet each player.

The 6 Key Players:

Player	Also Called	Simple Explanation
Customer	Cardholder	The person who buys something and pays with a card. The customer has a bank account and a card.
Merchant	Shop / Seller	The business that sells something. The merchant wants to receive money from the customer.
Acquiring Bank	Ekvaring / Merchant Bank	The merchant's bank. This bank receives the payment request and holds the merchant's money.
Issuing Bank	Emitent / Cardholder Bank	The customer's bank. This bank issued the customer's card and holds the customer's money.
Payment System	Payment Gateway / PSP	The company that connects the merchant to the banks. Like CIBPay. This is the "bridge."
Card Network	Visa / Mastercard / MilliKart	The network that connects acquiring banks and issuing banks. Like a "highway" between banks.

Let us look at a simple picture. The customer wants to buy a product from an online shop. The customer uses their card. The payment must travel from the customer's bank to the merchant's bank. But these two banks are different! They need a way to communicate. This is where the payment system and card network help.

Think of it like sending a letter: The customer puts a letter (payment request) in the mailbox. The payment system (CIBPay) picks it up and takes it to the post office (card network). The post office delivers it to the recipient's bank (issuing bank). The issuing bank checks: "Does this customer have enough money?" and sends back an answer: "Yes" or "No." The answer travels back the same way to the merchant.

CHAPTER 2

What is an Acquiring Bank (Ekvaring)?

The **acquiring bank** (also called "ekvaring bank" or "merchant bank") is the bank that works with the **merchant**. Every merchant that accepts card payments must have a contract with an acquiring bank. This bank provides the merchant with a "merchant account" where the payment money will be collected.

What does the acquiring bank do?

- **Opens a merchant account:** When a business wants to accept card payments, they sign a contract with an acquiring bank. The bank creates a special account called a "merchant account." All customer payments go into this account.
- **Receives payment requests:** When a customer pays, the payment request goes to the acquiring bank first. The acquiring bank checks: "Is this merchant active? Is the merchant account in good status?"

- **Forwards to the card network:** After the acquiring bank approves the merchant, it sends the payment request to the card network (Visa, Mastercard, MilliKart). The card network then sends it to the issuing bank.
- **Receives the money from issuing bank:** After the customer's bank (issuing bank) confirms the payment, the money eventually arrives at the acquiring bank's merchant account.
- **Sets up the merchant to the payment system:** The acquiring bank often works with a payment system like CIBPay to process transactions. The merchant connects to CIBPay, and CIBPay connects to the acquiring bank.
- **Charges fees:** The acquiring bank charges the merchant a fee for each transaction. This is usually a percentage of the transaction amount (for example, 1.5% - 3%). This is called the "merchant discount rate" or MDR.

Example in Azerbaijan: If an online shop signs a contract with Kapital Bank to accept card payments, then Kapital Bank is the acquiring bank (ekvaring) for this merchant. All customer payments will go through Kapital Bank before reaching the merchant.

CHAPTER 3

What is an Issuing Bank (Emitent)?

The **issuing bank** (also called "emitent bank" or "cardholder bank") is the bank that gave the card to the **customer**. When you go to a bank and ask for a debit card or credit card, that bank is your issuing bank. Your issuing bank holds your money and manages your card.

What does the issuing bank do?

- **Issues cards to customers:** The bank creates cards for its customers. Each card has a unique number, an expiry date, and a CVV code. The bank links the card to the customer's bank account.
- **Checks the customer's balance:** When a payment request arrives, the issuing bank checks: "Does this customer have enough money in their account?" If yes, the bank approves the payment. If no, the bank declines the payment.
- **Validates the card:** The bank checks: Is this card valid? Is it expired? Is it blocked? Is the CVV correct? If anything is wrong, the payment is declined.
- **Deducts money from the customer's account:** When the payment is approved, the bank deducts (removes) the money from the customer's bank account. The customer sees this as a charge on their bank statement.
- **Sends money to the acquiring bank:** Through the card network and the clearing/settlement process, the issuing bank sends the money to the acquiring bank where the merchant's account is.
- **Manages security:** The issuing bank monitors transactions for fraud. If the bank sees a suspicious transaction (for example, a payment in another country at 3 AM), it can block the transaction for security.

Example: If Zamir has a card from Pasha Bank, then Pasha Bank is his issuing bank. When Zamir buys something online, Pasha Bank checks if Zamir has enough money and deducts the amount from his account.

Important: The acquiring bank and issuing bank can be the SAME bank or DIFFERENT banks. If the customer and the merchant use the same bank, the process is simpler. But in most cases, they use different banks. This is where

the card network becomes important.

CHAPTER 4

What is a Payment System / Payment Gateway?

A **payment system** or **payment gateway** is a technology company that connects merchants to banks. It is like a "bridge" between the merchant's website and the banking system. CIBPay is a payment system. Other examples: PayTabs, Stripe, PayPal.

What does the payment system do?

- **Provides the API:** The payment system gives the merchant an API (a way to connect). The merchant's developer writes code that sends payment requests to the payment system's API.
- **Validates the merchant:** When a payment request arrives, the payment system first checks: "Is this merchant registered? Are the API credentials correct? Is the request format valid?"
- **Routes to the acquiring bank:** After validation, the payment system sends the request to the acquiring bank. The payment system knows which acquiring bank to use for each merchant.
- **Manages merchant accounts:** The payment system has a back-office (admin panel) where support specialists can create merchant accounts, manage API keys, check transaction history, and investigate problems.
- **Provides technical support:** When a merchant has a problem, they contact the payment system's support team. The IT Support Specialist helps the merchant fix integration issues, understand error codes, and resolve transaction problems.
- **Handles multiple payment methods:** The payment system can connect to multiple acquiring banks, card networks, and payment methods (Visa, Mastercard, MilliKart, digital wallets, etc.). The merchant does not need to integrate with each bank separately.

Think of the payment system as a translator: The merchant speaks "website language" and the bank speaks "banking language." The payment system translates between them. The merchant sends a simple API request, and the payment system converts it into the bank's required format. The bank sends back a response, and the payment system converts it into a simple format the merchant can understand.

CHAPTER 5

What is a Card Network?

A **card network** (also called "card scheme" or "payment network") is the organization that creates the rules and infrastructure for card payments. The most famous card networks are **Visa**, **Mastercard**, and in Azerbaijan, **MilliKart**.

What does the card network do?

- **Creates the rules:** The card network defines how card payments work. For example: maximum transaction amount, security requirements (like 3-D Secure), and how disputes (chargebacks) are handled.
- **Connects acquiring and issuing banks:** The card network is the "highway" between banks. The acquiring bank sends a message to the card network, and the card network routes it to the correct issuing bank. Think of it like a telephone network: you dial a number, and the network connects you to the right

person.

- **Operates the clearing and settlement system:** At the end of the day, the card network calculates how much each bank owes to other banks and transfers the money between them. This is called "clearing and settlement."
- **Provides security technology:** The card network provides security features like 3-D Secure (3DS), EMV chip technology, and fraud detection systems.

In Azerbaijan: Most local card payments use MilliKart. International cards use Visa or Mastercard. A payment system like CIBPay needs to be connected to all three networks to accept different types of cards.

CHAPTER 6

Step-by-Step: How Money Moves Between Banks

Now let us see the complete journey of a payment. We will follow the money from the customer's click to the actual money transfer. This is the most important chapter.

Part 1: Authorization (Real-Time, 1-3 seconds)

Authorization is when the system checks if the payment is possible. The money does NOT move yet. The system only checks: "Can we do this payment?" The customer sees "Payment Successful" but the merchant does not have the money yet.

Step 1: The **customer** clicks "Pay" on the merchant's website and enters their card details (card number, expiry date, CVV).

Step 2: The **merchant's website** sends a payment request to the **payment system** (CIBPay) via API. The request contains: amount, currency, card details, and merchant ID.

Step 3: The **payment system** (CIBPay) validates the request: checks merchant credentials, request format, and API key. If valid, CIBPay sends the request to the **acquiring bank**.

Step 4: The **acquiring bank** checks the merchant status. If the merchant is active and in good standing, the acquiring bank sends the request to the **card network** (Visa/Mastercard/MilliKart).

Step 5: The **card network** looks at the card number and identifies the **issuing bank**. For example, if the card starts with 4XXX, it is a Visa card issued by Pasha Bank. The card network routes the request to Pasha Bank.

Step 6: The **issuing bank** performs several checks: Is the card valid? Is it expired? Is the CVV correct? Does the customer have enough money? Is the transaction suspicious? The issuing bank sends back a response: "APPROVED" or "DECLINED" with a reason code.

Step 7: The response travels back: issuing bank -> card network -> acquiring bank -> payment system (CIBPay) -> merchant. The whole process takes 1-3 seconds.

Step 8: The **merchant** shows the result to the customer: "Payment Successful" or "Payment Failed." The customer receives an order confirmation.

IMPORTANT: At this point, the money has NOT moved yet! The customer's bank balance is "reserved" (held) but not yet deducted. The merchant does not have the money. This is only a promise that the payment will happen. The actual money movement happens later, during settlement.

Part 2: Clearing (Batch Process, End of Day)

Clearing is when banks exchange transaction information. At the end of each day, all the transactions from that day are collected and sent between the banks. Think of it like a teacher collecting all homework at the end of the day and distributing it to the correct graders.

Step 1: The **acquiring bank** collects all transactions from the day for all its merchants.

Step 2: The **card network** receives all transaction data from all acquiring banks.

Step 3: The **card network** sorts the transactions by issuing bank and sends each batch to the correct issuing bank. For example: all Pasha Bank card transactions go to Pasha Bank, all Kapital Bank card transactions go to Kapital Bank.

Step 4: The **issuing banks** receive the transaction lists and verify them against their own records. Now all banks agree on what happened today.

Part 3: Settlement (Actual Money Movement, T+1 to T+3 days)

Settlement is when the actual money changes hands. This usually happens 1-3 business days after the transaction. "T+1" means "transaction day plus 1 business day." This is the real money movement.

Step 1: The **issuing bank** deducts (removes) the money from the customer's account. The customer sees this as a completed charge on their bank statement.

Step 2: The **issuing bank** transfers the money to the **card network's settlement account**.

Step 3: The **card network** calculates net amounts: for example, if Pasha Bank's customers spent 100,000 AZN at merchants, and Pasha Bank's merchants received 80,000 AZN from customers, then Pasha Bank needs to pay 20,000 AZN to the card network.

Step 4: The **card network** transfers the money to the **acquiring bank**.

Step 5: The **acquiring bank** credits (adds) the money to the **merchant's account**. After deducting fees, the merchant receives the money. The merchant can now use this money.

Summary of the timeline: Authorization (1-3 seconds) -> Clearing (end of day) -> Settlement (T+1 to T+3 days). The customer sees the charge immediately, but the merchant receives the money 1-3 days later.

CHAPTER 7

Settlement and Clearing: The Real Money Movement

This chapter explains the difference between clearing and settlement in more detail. Many people confuse these two terms, but they are different.

Clearing = Exchanging Information

Clearing is when banks send transaction data to each other. No money moves during clearing. Banks only exchange information: "On Monday, your customer paid 100 AZN at my merchant." After clearing, all banks have the same list of transactions. This usually happens at the end of each business day.

Settlement = Moving Money

Settlement is when the actual money is transferred between banks. After clearing, the card network calculates the net amounts: how much each bank owes to other banks. Then the money is transferred. This usually happens 1-3 business days after the transaction.

Why does settlement take 1-3 days?

In the past, settlement took longer because banks had to physically process paper checks. Today, with electronic systems, it is faster, but there are still reasons for the delay: time zone differences, weekends, holidays, and the need for verification. In some modern systems, settlement can happen on the same day (real-time settlement).

Simple analogy: Clearing is like restaurants sharing orders at the end of the day: "My customers ordered 50 pizzas from your restaurant, and your customers ordered 30 pizzas from my restaurant." Settlement is when they actually pay each other the difference.

CHAPTER 8

Real Example: Customer Buys for 50 AZN

Let us follow a real example. Zamir buys a product for 50 AZN from an online shop using his Pasha Bank Visa card. The merchant uses CIBPay and their acquiring bank is Kapital Bank.

Authorization Phase (Real-Time):

Step	Who	What Happens
1	Zamir	Clicks "Pay" and enters Pasha Bank Visa card details on the shop website.
2	Merchant website	Sends API request to CIBPay: amount=50, currency=AZN, card=4XXX..., merchant_id=M123.
3	CIBPay	Validates the request (API key OK, format OK). Routes to acquiring bank (Kapital Bank).
4	Kapital Bank (Acquiring)	Checks merchant M123 status (active, OK). Sends to Visa card network.
5	Visa Network	Identifies the card as issued by Pasha Bank. Routes to Pasha Bank.
6	Pasha Bank (Issuing)	Checks: card valid, not expired, CVV OK, balance = 200 AZN (enough). Response: APPROVED.
7	Response back	APPROVED travels back: Pasha Bank -> Visa -> Kapital Bank -> CIBPay -> Merchant.
8	Merchant	Shows "Payment Successful" to Zamir. Order confirmed.

Settlement Phase (T+1 to T+3 days):

Step 1: Pasha Bank (issuing) deducts 50 AZN from Zamir's account. Zamir sees: "Online Shop -50 AZN" on his bank statement.

Step 2: Pasha Bank transfers 50 AZN to the Visa settlement account.

Step 3: Visa calculates net amounts for all banks and transfers money to **Kapital Bank** (acquiring). Visa deducts a small interchange fee (for example, 1-2%).

Step 4: Kapital Bank credits 50 AZN to the merchant's account, minus the acquiring bank fee (for example, 1%). The merchant receives approximately 49 AZN.

Step 5: If the merchant uses CIBPay, CIBPay may also deduct its own fee before the money reaches the merchant's final bank account.

Where does the money go? (Fee breakdown example for 50 AZN)

Fee Type	Who Charges	Amount	Explanation
Interchange Fee	Issuing Bank (Pasha)	~1.0 AZN	Fee paid by the acquiring bank to the issuing bank for processing the card payment.
Scheme Fee	Card Network (Visa)	~0.15 AZN	Fee paid to Visa for using their network.
Acquiring Fee	Acquiring Bank (Kapital)	~0.75 AZN	Acquiring bank's margin for providing the merchant account.
PSP Fee	Payment System (CIBPay)	~0.50 AZN	CIBPay's fee for providing the payment gateway service.
Merchant Receives		~47.60 AZN	The merchant receives the remaining amount after all fees.

CHAPTER 9

Other Important Workflows

9.1 Refund

A **refund** is when the merchant returns money to the customer. This happens when: the customer cancels an order, the product is defective, or the merchant made a mistake.

The process: The merchant sends a refund request to CIBPay. CIBPay sends it to the acquiring bank. The acquiring bank sends it through the card network to the issuing bank. The issuing bank credits (adds) the money back to the customer's account. The money is returned to the customer's card. This usually takes 3-7 business days.

Important: The refund goes back to the original card. You cannot refund to a different card. If the customer's card is expired or closed, the refund will fail.

9.2 Chargeback

A **chargeback** is when the customer contacts their bank and says: "I did not make this payment" or "I never received the product." The issuing bank opens a dispute and takes the money back from the merchant. This is different from a refund because the merchant does NOT agree to it.

The process: Customer contacts their issuing bank. The issuing bank creates a chargeback request. The money is taken from the merchant's acquiring bank account and returned to the customer. The merchant can dispute the chargeback by providing evidence (delivery confirmation, communication with customer, etc.).

9.3 3-D Secure (3DS)

3-D Secure (3DS) is an extra security layer for online payments. When the customer enters their card details, a popup appears from their bank asking them to enter a one-time password (OTP) sent to their phone. This proves that the real cardholder is making the payment.

The process: Customer enters card details. The payment system checks if 3DS is required. If yes, a redirect happens to the issuing bank's 3DS page. The customer enters their OTP. The bank confirms the identity and sends back an authentication result. The payment continues.

Benefit: If a 3DS-authenticated transaction is later disputed (chargeback), the merchant is protected because they have proof that the real cardholder made the payment.

9.4 Card Tokenization

Card tokenization is a security feature. Instead of storing the real card number, the system replaces it with a random string called a "token." For example, the real card number 4111-XXXX-XXXX-1234 is replaced with a token like tok_a8f3k2m9. This token can be used for future payments instead of the real card number.

The process: The customer enters their card details once. The payment system sends the card details to the card network or tokenization provider. The provider creates a token and returns it. Future payments use only the token. The real card number is never stored or transmitted again.

Use case: Tokenization is used for recurring payments (subscriptions), card-on-file (saving a card for future purchases), and payouts (sending money to customer cards). At Embafinans, we used PayTabs tokenization for credit disbursements to customer cards.

9.5 Payout / Disbursement

A **payout** (or disbursement) is when money is sent TO a customer's card or wallet. This is the opposite of a regular payment. Instead of the customer paying the merchant, the merchant (or system) pays the customer.

Card Payout: The system sends money to a customer's registered card. The process: The system uses a stored token (from tokenization) to identify the card. A payout request is sent through the payment system to the acquiring bank. The bank routes it through the card network to the issuing bank. The issuing bank credits the customer's account.

Wallet Payout: The system sends money directly to a customer's digital wallet (like Cuzdan) via API. This does not go through a bank or card network. The system sends an API request to the wallet provider. The wallet provider credits the customer's wallet balance. The wallet provider returns a success or failure response.

CHAPTER 10

CIBPay's Role in Detail

Now let us look at what CIBPay specifically does in the payment ecosystem. CIBPay is a **Payment Service Provider (PSP)**. This means CIBPay sits between the merchant and the banking system.

What CIBPay does:

- **Provides the API:** CIBPay gives merchants a REST API with endpoints for payments, refunds, voids, and status checks. The merchant's developer integrates with this API to accept payments on their website.
- **Test and production environments:** CIBPay provides a sandbox (test) environment where merchants can test their integration without real money. After testing, CIBPay moves the merchant to the production (live) environment.
- **Routes to the correct bank:** CIBPay knows which acquiring bank to use for each merchant. When a payment request comes in, CIBPay routes it to the correct bank and card network automatically.
- **Handles multiple payment methods:** CIBPay can process Visa, Mastercard, MilliKart, and potentially other methods like digital wallets. The merchant does not need separate integrations for each card type.
- **Manages webhooks:** After a payment, CIBPay sends a notification (webhook) to the merchant's server with the payment result. This tells the merchant: "Payment 12345 is successful, amount = 50 AZN."

- **Provides a back-office panel:** CIBPay has an admin panel where the support team can manage merchant accounts, check transactions, investigate problems, and configure settings.
- **Technical support:** CIBPay's IT Support team helps merchants with integration problems, error resolution, and payment investigations. This is YOUR role if you work at CIBPay.

Where CIBPay fits in the payment flow:

Customer -> Merchant Website -> **CIBPay (API)** -> Acquiring Bank -> Card Network -> Issuing Bank

CIBPay is the FIRST technology layer after the merchant. Everything that happens between the merchant and CIBPay is your responsibility. If a merchant has an API problem, an error they do not understand, or a transaction question, they come to you.

Your daily work at CIBPay would include:

1. Helping merchants integrate with CIBPay API (onboarding).
2. Troubleshooting API errors (checking logs, testing with Postman).
3. Investigating failed transactions (authorization declines, timeouts, network issues).
4. Managing merchant accounts in the back-office (credentials, webhooks, permissions).
5. Creating and maintaining documentation (API guides, error code references, FAQs).
6. Communicating with acquiring banks when there are bank-side issues.
7. Following up with merchants to ensure their problems are resolved.

CHAPTER 11

Interview Questions and Sample Answers

This chapter has interview questions related to the payment ecosystem with sample answers in simple English. Practice these answers before your CIBPay interview.

Q1: What is the difference between an acquiring bank and an issuing bank?

Sample Answer: The acquiring bank is the merchant's bank. It holds the merchant's account and receives payment requests. The issuing bank is the customer's bank. It issued the customer's card and holds the customer's money. When a payment happens, the acquiring bank receives the request and the issuing bank checks if the customer has enough money. They can be the same bank or different banks. In most cases, they are different banks, and the card network connects them.

Q2: Can you explain how money moves from one bank to another?

Sample Answer: Yes, there are three phases. First is authorization, which is real-time and takes 1-3 seconds. The customer pays, the merchant sends the request through the payment system to the acquiring bank, then to the card network, then to the issuing bank. The issuing bank checks the balance and responds with approved or declined. Second is clearing, which happens at the end of the day. Banks exchange transaction data through the card network. Third is settlement, which happens 1-3 business days later. The actual money moves from the issuing bank to the acquiring bank through the card network, and then to the merchant's account.

Q3: What role does a payment system like CIBPay play?

Sample Answer: CIBPay is the bridge between the merchant and the banking system. CIBPay provides the API that the merchant uses to accept payments. When a payment request comes in, CIBPay validates the merchant, checks the API credentials and request format, and then routes the request to the correct acquiring bank. CIBPay also provides a test environment for merchants to test their integration, a back-office panel for managing merchant accounts, webhook notifications for payment results, and technical support for integration issues.

Q4: What is card tokenization and why is it important?

Sample Answer: Card tokenization is a security feature. It replaces the real card number with a random token. For example, instead of storing 4111-XXXX-XXXX-1234, the system stores a token like tok_a8f3k2m9. The real card number is never stored or transmitted again. This is important for PCI-DSS compliance and security. I have hands-on experience with this. At Embafinans, I implemented a PayTabs card tokenization workflow for credit payouts to customer cards via Kapital Bank. The process was: register the card with a small verification amount, receive a token from PayTabs, and then use that token for all future payouts.

Q5: What is 3-D Secure and how does it work?

Sample Answer: 3-D Secure, or 3DS, is an extra security step for online payments. When the customer enters their card details, a popup appears from their bank asking them to enter a one-time password, or OTP, sent to their phone. This proves that the real cardholder is making the payment. The process is: the customer enters card details, the payment system checks if 3DS is required, if yes, a redirect happens to the issuing bank's 3DS page, the customer enters the OTP, the bank confirms the identity, and the payment continues. The benefit is that if a 3DS-authenticated transaction is later disputed, the merchant is protected because they have proof of the customer's identity.

Q6: What is the difference between clearing and settlement?

Sample Answer: Clearing is when banks exchange transaction information. It happens at the end of each business day. No money moves during clearing. Banks only share data about what transactions happened. Settlement is when the actual money is transferred between banks. This happens 1-3 business days after the transaction. The card network calculates the net amounts between banks and transfers the money. A simple analogy: clearing is like sharing a restaurant bill to see who ordered what, and settlement is when everyone actually pays their part.

Q7: A merchant says their payment is failing. How do you investigate?

Sample Answer: First, I ask the merchant for the error message and the transaction ID. Then I check where in the payment flow the failure happened. Did the request reach CIBPay? If not, it is a merchant-side issue - maybe their code or network. If the request reached CIBPay, did CIBPay send it to the acquiring bank? If the bank declined it, I check the decline reason code. Common reasons: insufficient funds, expired card, incorrect CVV, card blocked. I explain the reason to the merchant in simple language and advise them on what to do. If it is a server-side error, I escalate to our development team. I also have experience investigating deeper issues. At Embafinans, when card tokenization payouts failed, I analyzed application and API logs, identified the error, and coordinated with both Kapital Bank support and PayTabs support to resolve the issue.

Final Tip: In the interview, when they ask about payment workflows, always mention that you understand the COMPLETE flow: authorization (real-time) -> clearing (end of day) -> settlement (T+1 to T+3). This shows you understand not just the API layer, but the actual banking process behind it. Also mention your hands-on experience with card tokenization and wallet integration - this proves you have practical knowledge, not just theory.